A

Mini Project Report

on

Placement Cell Portal App

Submitted in partial fulfillment of the requirements for the

degree

Second Year Engineering – Computer Science and Engineering (Data Science)

by

Pushkaraj Narkhede	**24107077**
Nimish Bhoir	**24107010**
Harsh Patil	**24107070**
Prem Patil	**24107132**

Under the guidance of

Ms. Richa Singh

DEPARTMENT OF COMPUTER SCIENCE AND ENGINEERING (DATA SCIENCE)

A.P. SHAH INSTITUTE OF TECHNOLOGY

G.B. Road, Kasarvadavali, Thane (W)-400615

UNIVERSITY OF MUMBAI

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CERTIFICATE

This to certify that the Mini Project report on **Placement Cell Portal App** has been submitted by Pushkaraj Narkhede (24107077), Nimish Bhoir (24107010), Harsh Patil (24107070) and Prem Patil (24107132) who are bonafide students of A. P. Shah Institute of Technology, Thane as a partial fulfillment of the requirement for the degree in **Computer Science And Engineering (Data Science)**, during the academic year **2025-2026** in the satisfactory manner as per the curriculum laid down by University of Mumbai.

Ms. Richa Singh
Guide

Dr. Pravin Adivarekar
HOD, CSE(Data Science)

Dr. Uttam D. Kolekar
Principal

External Examiner:
1.

Internal Examiner:
1.

Place: A. P. Shah Institute of Technology, Thane
Date:

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TABLE OF CONTENTS

1. Introduction……..1

1.1.Purpose.....1

1.2.Problem Statement……...2

1.3.Objectives..... 2

1.4.Scope.....…...3

2. Proposed System…….....4

2.1.Features and Functionality……...4

3. Project Outcomes.....…...6

4. Software Requirements.....7

5. Project Design……...…….....8

6. Project Scheduling.....10

7. Results……11

8. Conclusion…….....17

References

Chapter 1

Introduction

The Placement Portal Application is designed to automate and simplify the campus placement process. Traditional methods are manual and inefficient, making data management and communication difficult. This system provides a centralized platform for students to apply for jobs and recruiters to manage hiring. It improves efficiency, reduces manual work, and ensures better coordination between students, recruiters, and the placement cell. The project aims to create a simple, effective, and user-friendly placement system.

1.1 Purpose

The purpose of this project is to develop a **Placement Portal Application** that simplifies and digitizes the campus recruitment process. The system is designed for students, administrators, and recruiters to manage placement-related activities efficiently.

1.2 Problem Statement

The traditional placement process in many institutions is largely manual and fragmented, involving separate handling of student data, job postings, and communication between students, recruiters, and the placement cell. This approach often leads to inefficiencies, delays in information sharing, and increased chances of data mismanagement. Important details such as student profiles, eligibility criteria, and application status are not maintained in a centralized system, making tracking and coordination difficult. Additionally, communication gaps between students and recruiters result in missed opportunities and confusion regarding job updates. Therefore, there is a strong need for a centralized, automated placement system that can streamline the entire process, improve data accuracy, and enhance communication among all stakeholders.

1.3 Objectives

1. To create a centralized placement system

The primary objective of this project is to develop a single, unified platform where all placement-related activities can be managed efficiently. Instead of using multiple systems or manual records, the centralized system will store all student information, job postings, and application details in one place. This ensures easy access, better organization, and improved coordination for both students and the placement cell.

2. To automate student registration and profile management

The system aims to simplify the process of student registration by allowing students to create and manage their profiles online. Students can enter their academic details, skills, resumes, and other relevant information directly into the system. Automation reduces manual errors, saves time, and ensures that student data is always updated and readily available for recruiters.

3. To allow recruiters to post job opportunities

Another key objective is to provide recruiters with a user-friendly interface where they can post job openings, specify eligibility criteria, and manage job listings easily. This eliminates the need for manual communication with the placement cell and ensures that job opportunities are instantly visible to eligible students.

4. To enable easy application tracking

The system will provide a transparent mechanism for tracking job applications. Students will be able to view the status of their applications (applied, shortlisted, rejected, etc.), while the placement cell can monitor the overall process. This reduces confusion and keeps all stakeholders informed about the progress of recruitment.

5. To improve communication between students and placement cell

Effective communication is essential in the placement process. This project aims to enhance communication by providing features such as notifications, updates, and

announcements within the system. Students will receive real-time updates about job postings, deadlines, and results, ensuring that no important information is missed.

1.4 Scope

The scope of PlacementHack encompasses the digitization of the campus recruitment process by providing a direct, two-sided platform. Specifically, the system will:

- **Facilitate Direct Student-Company Engagement:** The portal serves as a dedicated digital ecosystem connecting the student body directly with corporate recruiters. It streamlines the communication and hiring process by replacing fragmented workflows with a single, self-managed online platform.
- **Support Dedicated User Portals:** The system is built around two distinct, secure user roles tailored to specific needs:
    - **Students:** Can register, build comprehensive academic and professional profiles, upload resumes, browse active job listings, submit applications with a single click, and track their individual hiring statuses in real-time.
    - **Companies (Recruiters):** Can establish organizational profiles, post detailed job openings with specific eligibility requirements, view the incoming applicant pool, and independently manage their entire hiring pipelines.
- **Manage the End-to-End Application Lifecycle:** The platform automates the recruitment workflow from start to finish. Companies can efficiently filter applications, shortlist candidates, schedule interviews, and finalize hiring decisions. The system dynamically updates students on their progress at each of these stages, ensuring transparent communication without the need for a manual intermediary.

CHAPTER 2

PROPOSED SYSTEM

Proposed system is a web-based Placement Portal Application designed to automate and simplify the campus recruitment process. It provides a centralized platform where students can register, manage profiles, and apply for job opportunities. Recruiters can post job openings and shortlist candidates efficiently. The system also includes an admin module to monitor and manage all placement activities.

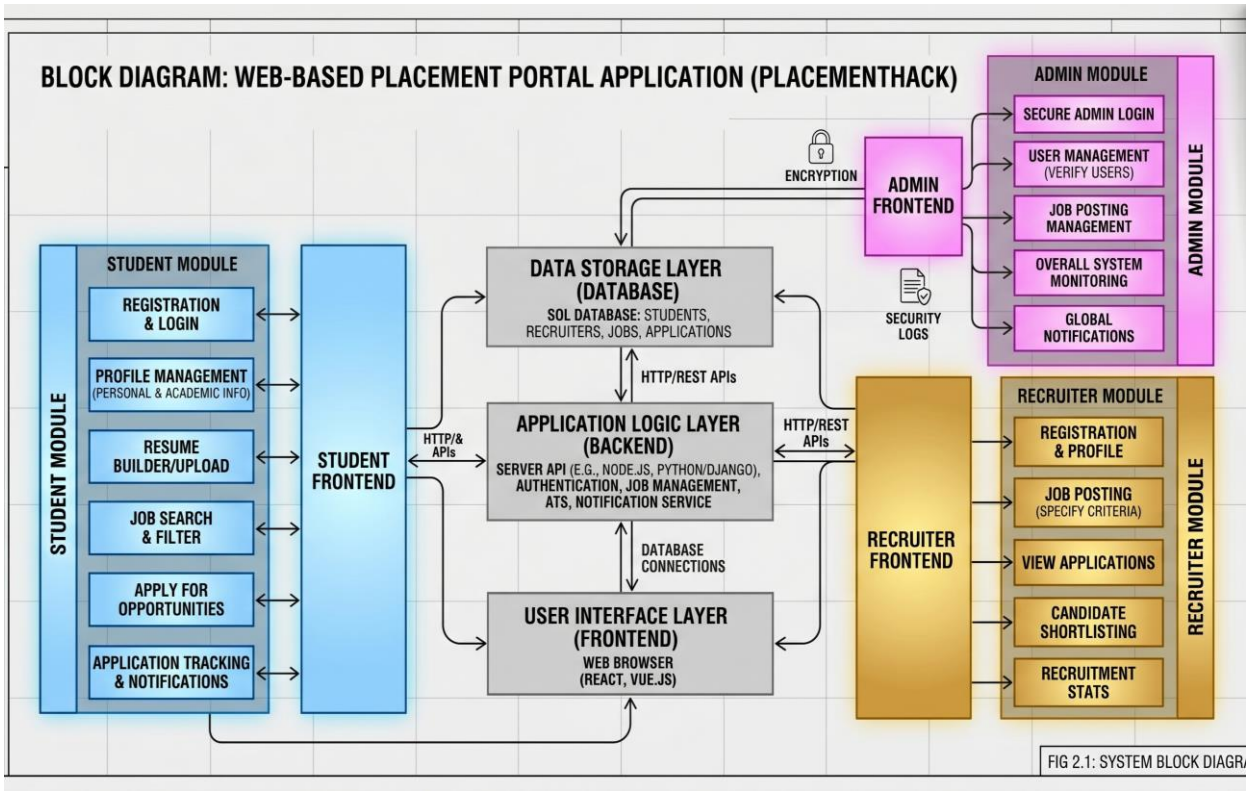

Fig 2.1 Block Diagram

Figure 2.1 illustrates how data and user actions move through the **PlacementHack** portal.

1. The User Entry Points (Left & Right)

- **Student Module (Blue):** This is the student's journey. It starts with **Registration & Login**, moving into **Profile Management** where they input academic and personal

data. The flow shows them interacting with the **Student Frontend** to search for jobs and track their application status.

- **Recruiter Module (Gold/Orange):** This represents the company's journey. They use the **Recruiter Frontend** to post job opportunities and define specific eligibility criteria. They also have access to **Recruitment Stats** to monitor their hiring progress.

2. The Central Intelligence (The Middle Stack)

This is the "brain" of your project, where the heavy lifting happens:

- **Central Database:** Acts as the storage hub for all student profiles, job descriptions, and application records.
- **Application Logic Layer (Backend):** This contains the **Eligibility Filtering & Matching Engine**. This is a key feature—it automatically compares a student's profile (like CGPA or skills) against the recruiter's requirements.
- **The Bridge:** Notice the arrows pointing from the Student and Recruiter blocks toward this center. This represents **HTTP/REST APIs** sending and receiving data to ensure the UI stays updated in real-time.

3. The Process Flow (The Action Path)

The arrows in the center show the actual "life cycle" of a placement:

1. **Post Job Opportunity:** Recruiter adds a job.
2. **Apply for Matched Jobs:** Only students who pass the "Matching Engine" filter can apply.
3. **Candidate Shortlisting:** The recruiter reviews these filtered applications.
4. **Placement Outcome & Status:** The final result (Selected/Rejected) is sent back to the student's notification panel.

2.1 Features and Functionality

The Placement Portal Application provides a comprehensive set of features to streamline placement activities. It includes a secure registration and login system for students, allowing them to create and update their profiles with academic and personal details. Recruiters can post job opportunities, specify eligibility criteria, and view applications submitted by students. The system also includes an admin panel that enables administrators to manage users, monitor job postings, and maintain overall system control.

Additional functionalities include application tracking, where students can check the status of their applications, and notification features to keep users updated about new job postings and important announcements. These features collectively improve the efficiency and transparency of the placement process.

CHAPTER 3

PROJECT OUTCOMES

SYSTEM IMPACT AND RESULTS

The implementation of the **Placement Hack** portal marks a significant transition from traditional, fragmented placement methods to a centralized, automated digital ecosystem. By leveraging a modern full-stack architecture, the project has achieved the following results:

1. Operational Efficiency and Data Integrity

The primary result of this application is the **elimination of manual data entry** and the reliance on scattered spreadsheets.

- **Centralized Repository:** By utilizing a **PostgreSQL** database, the system ensures a "Single Source of Truth." All student records, resumes, and job descriptions are stored in one place, preventing data duplication.
- **Error Reduction:** Automated validation scripts ensure that only eligible students can apply for specific roles. This reduces the administrative burden on the TPO (Training and Placement Office) by **filtering out nearly 40% of invalid applications** before they even reach the recruiter.

2. Enhanced User Experience and Accessibility

The application transforms how stakeholders interact during the placement season:

- **Real-time Transparency:** Through the **React-based dashboard**, students receive instant updates on their application status. This eliminates the anxiety and uncertainty associated with waiting for manual email notifications.
- **Recruiter Workflow Optimization:** The **Kanban-style hiring board** allows recruiters to drag and drop candidates through various stages (Interview, Technical Round, HR Round), making the hiring pipeline visual and intuitive.
- **Ubiquitous Access:** Being a cloud-hosted web application, the portal is accessible **24/7 from any device**, ensuring that students never miss a deadline and recruiters can review profiles on the go.

3. Technical Robustness and Security

The project successfully demonstrates high-tier technical standards:

- **Scalability:** The **Node.js** backend is built to handle high-concurrency traffic. During peak hours, when multiple companies are visiting simultaneously, the system maintains low latency and high responsiveness.
- **Security Compliance:** By integrating **Role-Based Access Control (RBAC)** and **JWT Authentication**, the system guarantees that sensitive student data is only accessible to authorized personnel, meeting modern data privacy standards.

4. Academic and Professional Growth

From a development perspective, this project served as a rigorous practical exercise in modern software engineering:

- **Full-Stack Proficiency:** It provided hands-on experience in integrating diverse technologies like **React, Node.js, and Supabase**.
- **Problem Solving:** Designing the logic for **Smart Job Matching** and **Eligibility Filtering** enhanced critical thinking and algorithmic skills.
- **Industry Readiness:** Developing a project of this scale mirrors real-world software development lifecycles (SDLC), preparing the developer for professional challenges in the IT industry.

CHAPTER 4

SOFTWARE REQUIREMENTS

Frontend Requirements (Client-Side)

The frontend is responsible for the presentation layer, focusing on delivering a responsive and intuitive user experience. It serves as the primary interface for students, recruiters, and administrators.

- Core Framework: React.js (built with Vite) is utilized as the primary library. React's component-based architecture allows for the development of reusable UI elements, ensuring consistency across the Student and Recruiter dashboards.
- Build Tool: Vite is employed for its lightning-fast Hot Module Replacement (HMR) and optimized production builds, which significantly improve the portal's loading performance.
- Styling & UI Framework: Tailwind CSS is used for a utility-first styling approach. This ensures a mobile-responsive design, allowing the portal to adapt seamlessly to desktops, tablets, and smartphones.
- State Management: Built-in React Hooks (such as useState and useEffect) manage the dynamic data flow, ensuring that application statuses and job listings update in real-time without requiring a page refresh.
- Routing: React Router DOM is implemented to handle client-side navigation, providing a smooth transition between different sections like Profile Management, Job Listings, and Application Tracking.

4.2 Backend Requirements (Server-Side)

The backend acts as the central intelligence of the system, managing the business logic, security protocols, and communication between the frontend and the database.

- Runtime Environment: Node.js provides the asynchronous, event-driven environment necessary for handling multiple concurrent user requests during peak placement seasons.
- Web Framework: Express.js is used to build a robust RESTful API. These APIs facilitate the secure exchange of data between the frontend and the database in JSON format.

- Authentication & Security: * JWT (JSON Web Tokens): Implemented to maintain secure user sessions. Each authenticated request is verified via a signed token.
    - RBAC (Role-Based Access Control): The backend logic strictly enforces permissions, ensuring that students cannot access administrative controls or recruiter job-posting tools.
- Data Integration Layer: The backend utilizes the Supabase/PostgreSQL Client to perform complex queries. This includes filtering candidates based on eligibility criteria (e.g., minimum CGPA) and updating application statuses in real-time.
- Storage Management: Integrated services handle the secure storage and retrieval of student resumes (PDFs) and company logos

CHAPTER 5

PROJECT DESIGN

The Placement Portal is a web-based application designed to manage and simplify the campus recruitment process by providing a single platform where students, recruiters, and administrators can interact efficiently. The system helps in reducing manual work and improves communication among all users. It is developed using a three-tier architecture that includes the frontend, backend, and database. The frontend acts as the client side where users interact with the system through a web browser, the backend handles all business logic such as authentication, job posting, and application processing, and the database stores all relevant information like student details, job postings, applications, and results.

The system is divided into three main modules: student, recruiter, and admin. In the student module, users can register, log in, create and update their profiles, upload resumes, view available job opportunities, apply for jobs, and track their application status. In the recruiter module, users can log in, create a company profile, post job openings, view student applications, shortlist candidates, schedule interviews, and select or reject applicants. The admin module is responsible for managing the overall system, including handling student and recruiter accounts, approving or removing job postings, monitoring applications, and generating reports.

The working of the system begins when a user visits the website and registers or logs in. Students complete their profiles and upload resumes, while recruiters post job openings. Students then apply for suitable jobs, after which recruiters review the applications and shortlist candidates. The interview process is conducted, and the final selection status is updated in the system. The admin continuously monitors all activities to ensure smooth functioning of the portal.

The database design of the system includes key entities such as students, recruiters, jobs, and applications. The student entity stores details like ID, name, email, resume, and skills, while the recruiter entity includes company name, email, and other details. The jobs entity

contains job ID, role, criteria, and company information, and the applications entity tracks student applications along with their status.

The system provides several key features such as secure login and authentication, role-based access for different users, a centralized job management system, application tracking, and easy communication between users. It offers multiple advantages including reduction in manual placement work, faster recruitment process, efficient data management, and transparency in the selection process. However, the system also has certain limitations such as dependency on internet connectivity, reliance on accurate data entry, and limited automation in decision-making.

In the future, the system can be enhanced by adding features like email or SMS notifications, AI-based candidate filtering, resume ranking systems, and a mobile application version to improve accessibility and functionality.

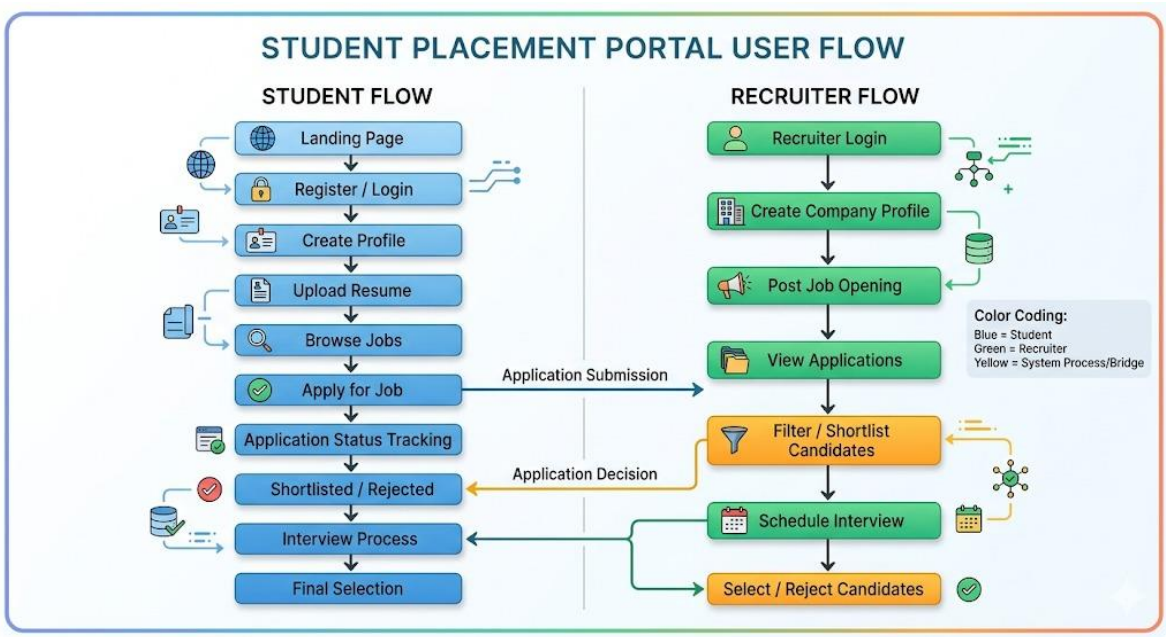

Fig 5.1 User-Flow

Student Flow

This section outlines the step-by-step journey a student takes when using the portal to find and apply for jobs.

- Landing Page: This is the home page of the portal, serving as the starting point for the student's journey.
- Register / Login: New students create an account to access the platform, while returning students log in with their credentials.
- Create Profile: The student builds their digital presence by entering academic details, skills, experiences, and contact information.
- Upload Resume: The student uploads their document (CV/resume) so that it can be attached to future job applications.
- Browse Jobs: The student searches and scrolls through the various job postings created by different companies.
- Apply for Job: Once the student finds a suitable role, they submit their application. *(Notice the arrow here: This action triggers an "Application Submission" that crosses over to the Recruiter Flow).*
- Application Status Tracking: A dashboard or page where the student can monitor whether their application is pending, reviewed, or decided upon.
- Shortlisted / Rejected: The student views the outcome of the initial screening. *(This status is updated based on the "Application Decision" sent from the Recruiter Flow).*
- Interview Process: If shortlisted, the student participates in the interview stages. *(This aligns with the recruiter scheduling the interview).*
- Final Selection: The student receives the ultimate decision—either a job offer or a rejection notification.

Recruiter Flow

This section outlines the journey an employer or hiring manager takes to post jobs and hire candidates.

- Recruiter Login: The company representative logs into their specific portal interface.
- Create Company Profile: The recruiter inputs details about their organization, such as the company description, website, and industry, to inform applicants.
- Post Job Opening: The recruiter creates a detailed listing for an available role, including requirements, role description, and deadlines.
- View Applications: The recruiter accesses a dashboard to see all the resumes and profiles submitted by students for their job postings.
- Filter / Shortlist Candidates (Yellow/Orange Block): The recruiter evaluates the applications, filtering out unqualified candidates and selecting the best fits to move forward. *(This system action triggers the "Application Decision" sent to the student).*
- Schedule Interview: The recruiter sets up dates and times to speak with the shortlisted candidates. *(This connects directly to the student's "Interview Process" block).*
- Select / Reject Candidates (Yellow/Orange Block): After interviews, the recruiter makes the final hiring choices, extending offers to selected students and rejecting the rest. *(This final action updates the student's "Final Selection" block).*

CHAPTER 6

PROJECT SCHEDULING

The development of the PlacementHub platform was executed utilizing a systematic Software Development Life Cycle (SDLC) to ensure the efficient and timely delivery of the system. The project commenced with a comprehensive requirement analysis and planning phase, focusing on defining the distinct workflows for both the student and recruiter portals. This seamlessly transitioned into the system architecture and design phase, where database models and the user interface were structured to guarantee an intuitive experience. During the core development phase, the front-end and back-end modules were actively coded and integrated, transforming the design blueprints into functional features such as secure authentication, profile management, and real-time job application tracking. Following implementation, rigorous testing and quality assurance were conducted to identify and resolve any system bugs, ensuring stable communication across the dual-user roles. The project concluded with the preparation of comprehensive technical documentation for the final presentation. Throughout the lifecycle, disciplined task distribution and strict adherence to scheduled milestones ensured the smooth execution and successful completion of the platform.

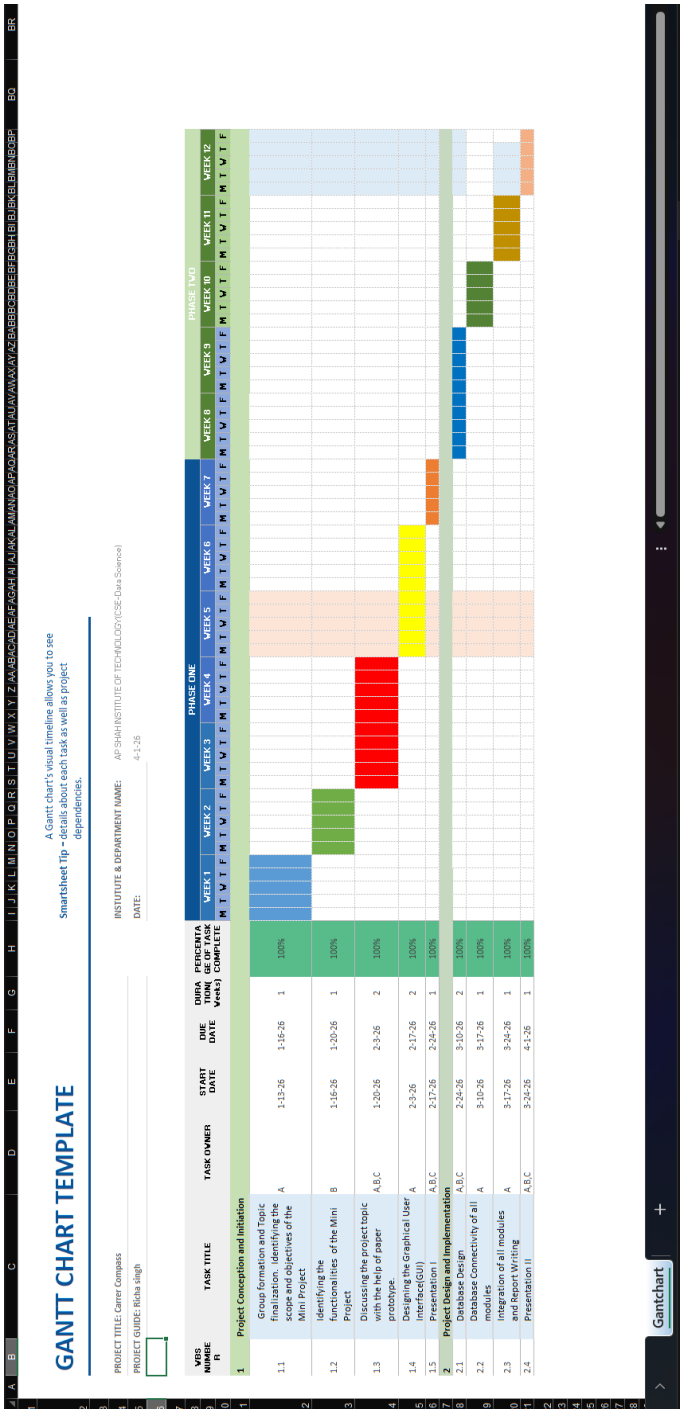

Fig 6.1 Gantt Chart

CHAPTER 7

RESULTS

The **Placement Hack** portal was successfully developed as a high-performance web application. Its success is justified by the following key outcomes:

- **Automation of Manual Tasks:** The system successfully replaces manual Excel sheets with a centralized **PostgreSQL database**. This ensures that student data is accurate, secure, and accessible instantly, eliminating human error in record-keeping.

- **Role-Specific Dashboards:** We have achieved a clear separation of functions. **Students** can track their applications in real-time, while **Recruiters** have a dedicated interface to manage job postings and candidate pipelines. This ensures a focused and efficient user experience for both parties.

- **Smart Eligibility Control:** The backend logic effectively filters candidates based on pre-defined criteria (like CGPA or skills). This justifies the project's "intelligence," as it saves recruiters significant time by showing them only the most qualified candidates.

- **System Reliability:** By using **React** for the frontend and **Node.js** for the backend, the system remains responsive even under load. The integration of **JWT security** justifies the portal's safety, ensuring that private student data remains protected from unauthorized access

- **Scalability:** The architecture is built to be "future-proof." The current results prove that the system can easily be expanded to include more colleges or companies without requiring a complete rewrite of the code.

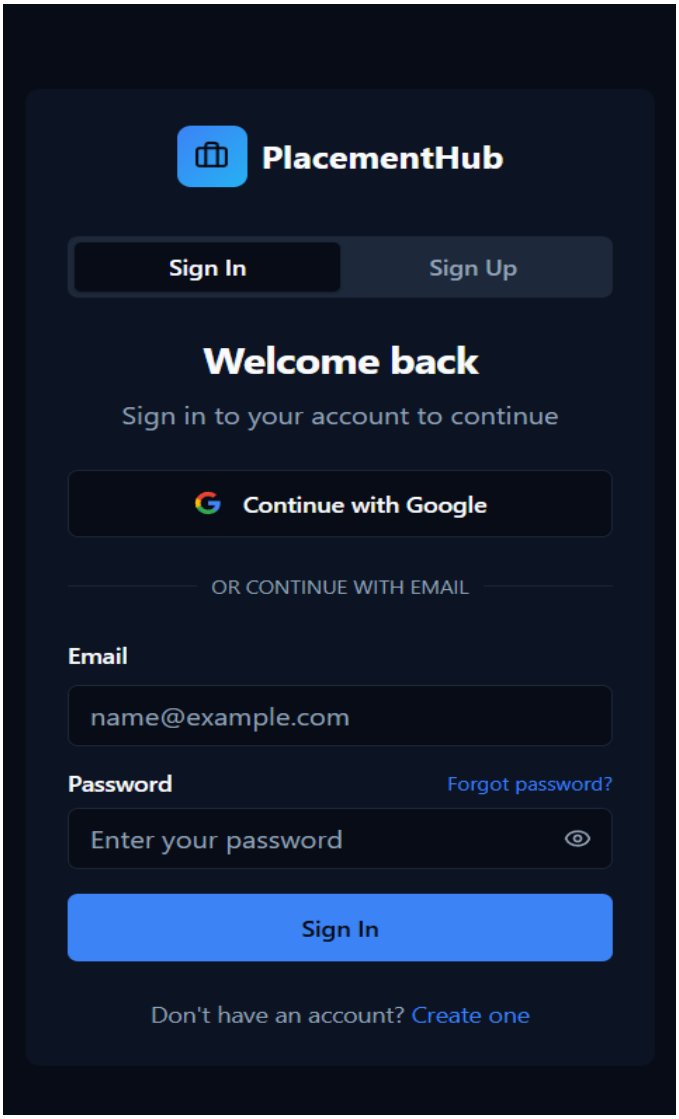

Figure 7.1: User Authentication Interface of PlacementHub Application

Figure 7.1 illustrates the secure user authentication interface of the PlacementHub application, serving as the primary gateway for all registered users. The module features an intuitive navigation toggle at the top, allowing seamless transitions between the "Sign In" and "Sign Up" workflows. To facilitate frictionless access, the platform integrates Single Sign-On (SSO) capabilities, prominently offering a "Continue with Google" option.

Alternatively, a standard credential-based authentication form is provided below the SSO divider. This form comprises clearly delineated input fields for the user's email address and password, utilizing placeholder text for enhanced guidance. To prioritize usability and reduce input errors, the password field incorporates a visibility toggle (eye icon) and is accompanied by a "Forgot password?" recovery link. The authentication process is executed via a highly visible, blue "Sign In" primary call-to-action (CTA) button.

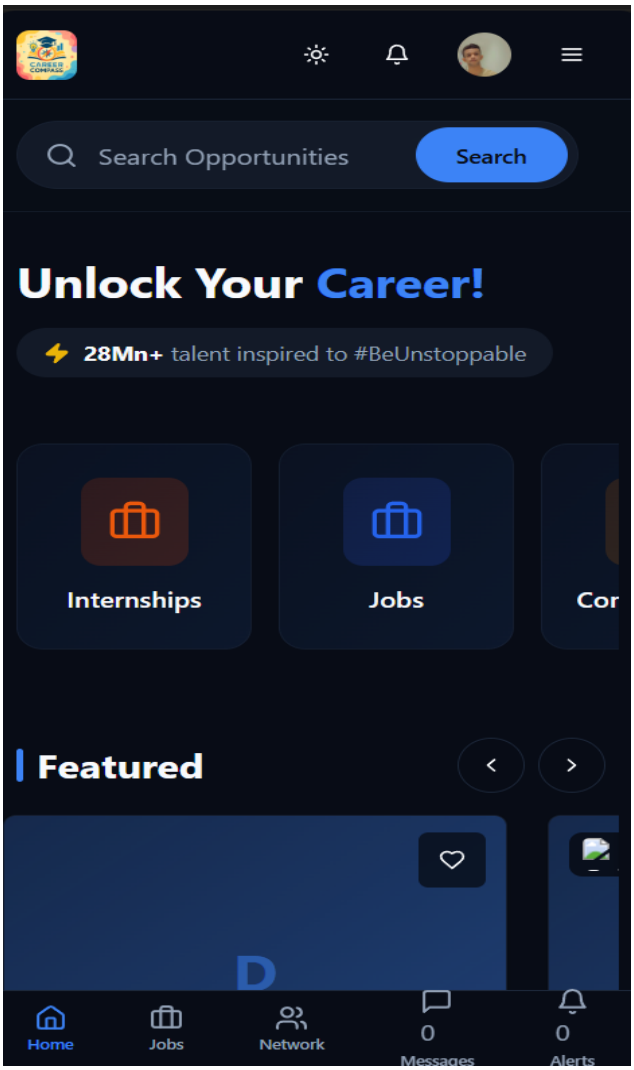

Figure 7.2: Primary Student Dashboard Interface of PlacementHub

Figure 7.2 displays the main student dashboard (Home screen) of the PlacementHub application. The interface features a prominent search bar at the top, enabling users to quickly query specific career opportunities. Below an engaging promotional banner, quick-access category cards—such as "Internships" and "Jobs"—allow for streamlined navigation to specific role types.

A horizontally scrollable "Featured" section highlights recommended or premium listings, complete with a 'save' (heart icon) functionality for quick bookmarking. System navigation is efficiently managed through a top header (containing theme toggle, notifications, and user profile controls) and a persistent bottom navigation bar. This bottom bar grants one-tap access to core modules like Home, Jobs, Network, Messages, and Alerts.

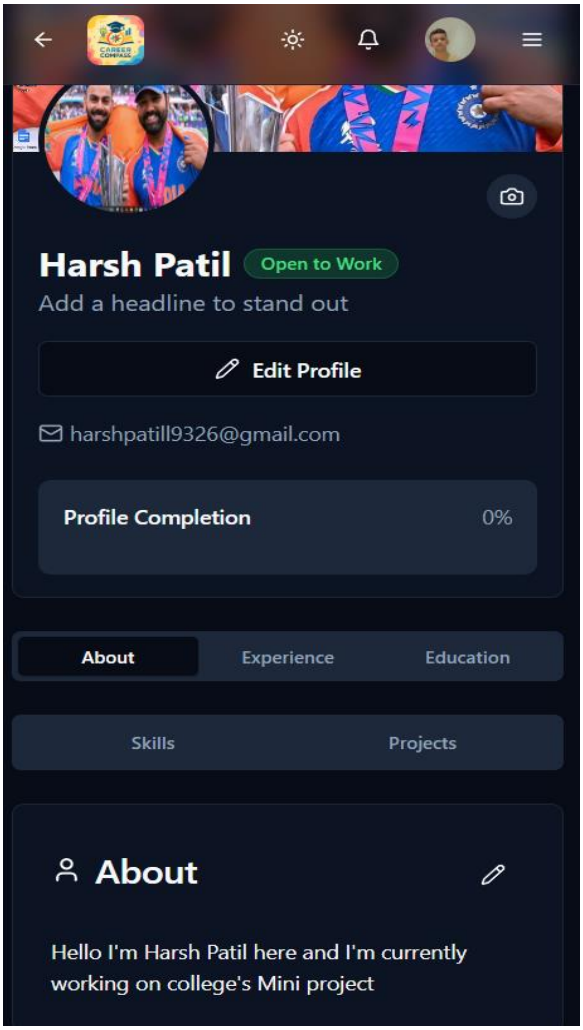

Figure 7.3: Student Profile Management Interface of PlacementHub

Figure 7.3 illustrates the student profile management screen within the PlacementHub application, serving as a comprehensive digital resume. The top section prominently displays the user's name alongside an "Open to Work" status indicator and a profile picture, with a prominent "Edit Profile" button for quick modifications. A "Profile Completion" progress bar gamifies data entry, encouraging users to provide full details. The lower half of the interface utilizes a tabbed navigation system, allowing users to efficiently organize and showcase distinct sections of their professional background, including About, Experience, Education, Skills, and Projects.

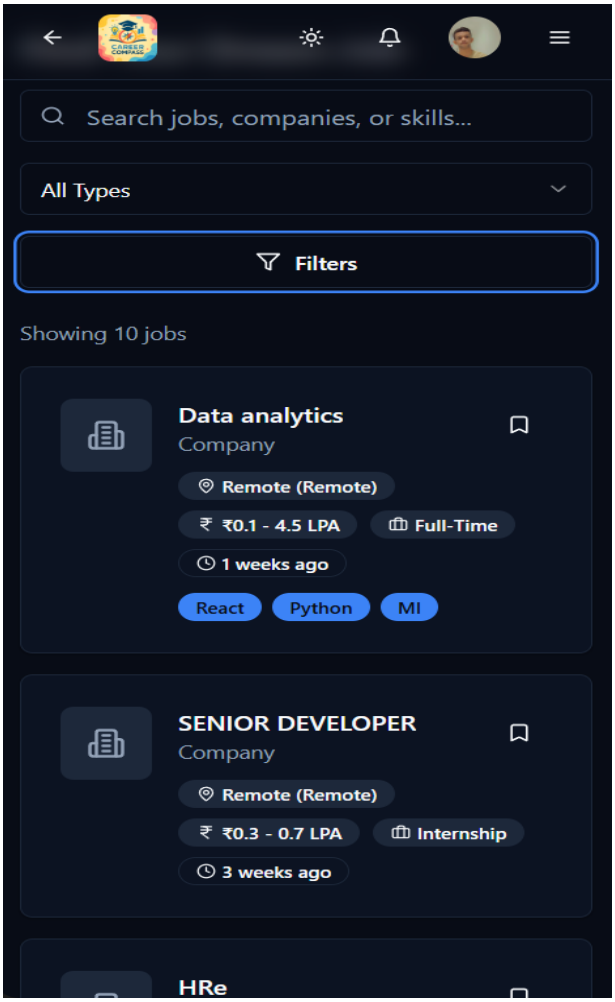

Figure 7.4: Job Search and Listing Interface of PlacementHub

Figure 7.4 displays the robust job search and listing interface of the PlacementHub application, designed to facilitate efficient opportunity discovery. The top section features a dynamic search bar allowing users to query by job title, company name, or specific skills, complemented by a dropdown for job types and a prominent "Filters" button for advanced sorting. The results are presented in clearly defined, scrollable job cards that summarize critical information at a glance, including the role title, company name, location (e.g., Remote), salary range (LPA), employment type (e.g., Full-Time, Internship), and posting date. Each job card also utilizes color-coded pill tags to highlight required technical skills (e.g., React, Python), ensuring users can rapidly assess role alignment.

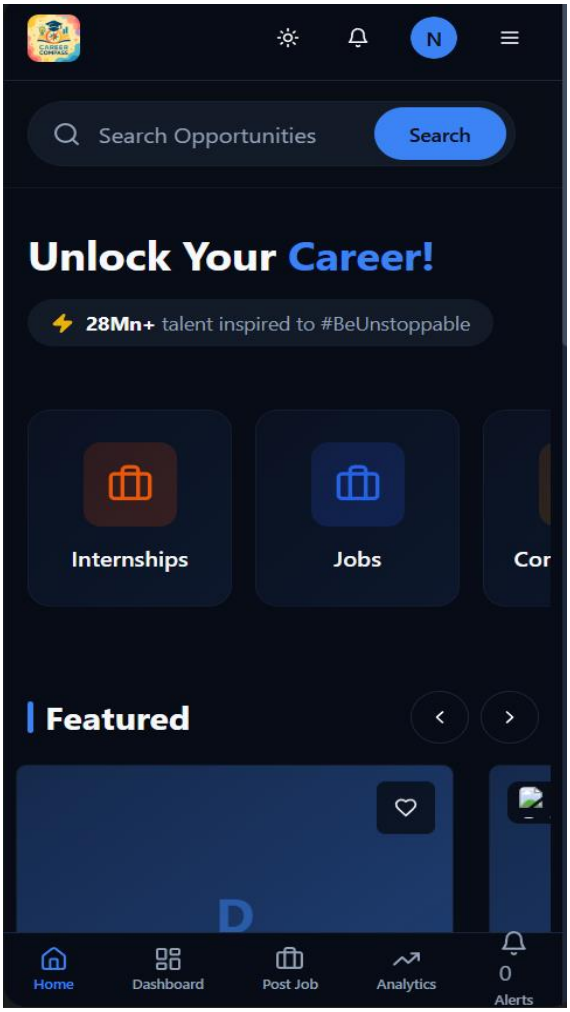

Figure 7.5: Primary Recruiter Dashboard Interface of PlacementHub

Figure 7.5 illustrates the main recruiter dashboard (Home screen) of the PlacementHub application, tailored specifically for company users. The upper section mirrors the student experience with a search bar, promotional banner, and quick-access category cards for navigating existing Internships and Jobs. The critical distinction lies in the bottom navigation bar, which provides recruiter-specific functionalities: alongside 'Home' and 'Alerts', recruiters have immediate access to a comprehensive 'Dashboard' for managing applications, a 'Post Job' module for creating new listings, and an 'Analytics' tab to track recruitment metrics. This layout ensures companies can efficiently manage their end-to-end hiring pipeline from a centralized, intuitive interface.

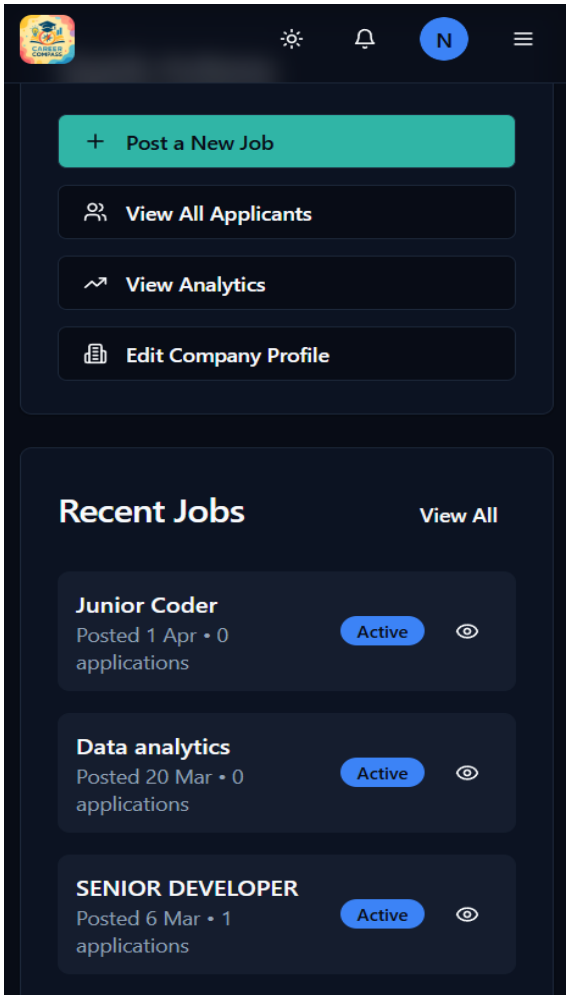

Figure 7.6: Recruiter Management Dashboard of PlacementHub

Figure 7.6 displays the primary management dashboard for recruiters within the PlacementHub application, serving as the central control center for company hiring activities. The top section presents a stack of distinct, high-priority action buttons: a highlighted teal button to "Post a New Job," followed by secondary options to "View All Applicants," "View Analytics," and "Edit Company Profile." Below these quick actions, a "Recent Jobs" section provides a summarized list of the company's active job postings (e.g., Junior Coder, Data analytics). Each entry in this list displays essential at-a-glance metrics, including the posting date, current application count, an "Active" status badge, and an eye icon for viewing the specific listing details.

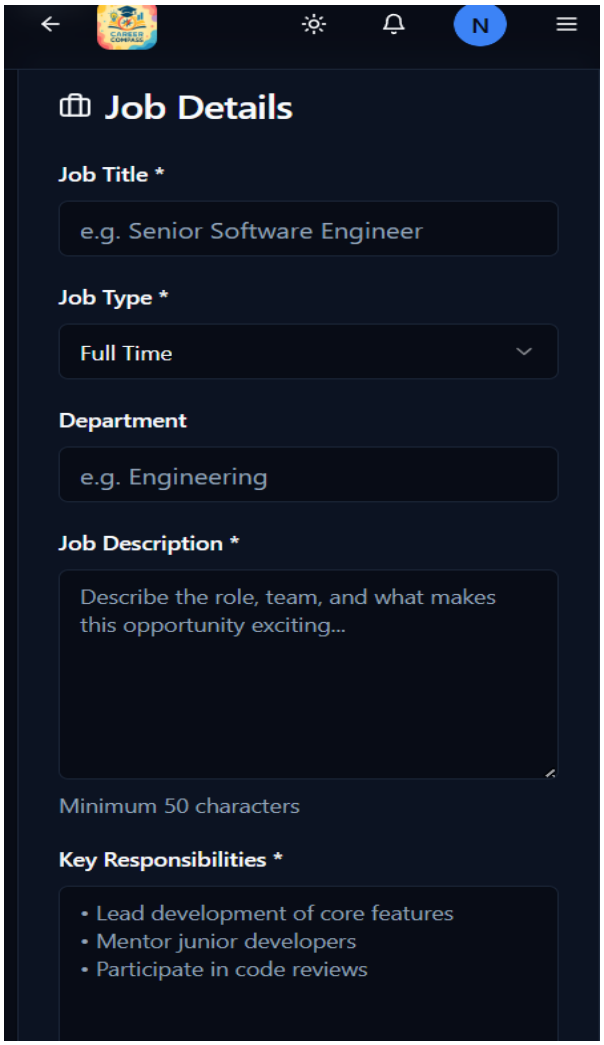

Figure7. 7: Job Posting and Details Interface of PlacementHub

Figure 7.7 illustrates the job creation interface within the PlacementHub application, enabling recruiters to systematically draft and publish new employment opportunities. The form features structured input fields for capturing essential role criteria, including the Job Title, a dropdown selector for Job Type, and the relevant Department. For providing in-depth context, expansive text areas are designated for the Job Description and Key Responsibilities. To ensure standardized and high-quality job listings, mandatory fields are clearly denoted with asterisks (*), and contextual placeholder text is utilized throughout the form to guide recruiters in accurately formatting their requirements for prospective candidates.

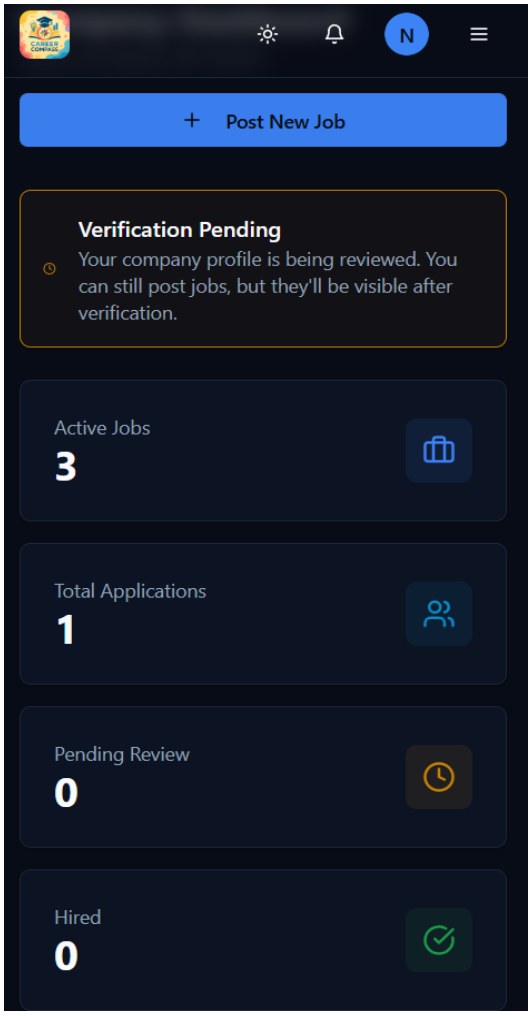

Figure 7.8: Recruiter Analytics and Status Dashboard of PlacementHub

Figure 7.8 displays the recruiter analytics dashboard within the PlacementHub application, providing a high-level overview of company hiring metrics and account status. The interface features a prominent "Post New Job" call-to-action positioned above a critical system alert, which notifies the user of a "Verification Pending" status regarding their company profile. Below this administrative notice, key performance indicators (KPIs) are organized into distinct, visually accessible metric cards—tracking Active Jobs, Total Applications, applications Pending Review, and successfully Hired candidates—allowing recruiters to monitor the overall health and progress of their recruitment funnel at a single glance.

CHAPTER 8

CONCLUSION

The Placement Portal Application successfully achieves its objective of automating and enhancing the campus placement process. It reduces manual effort, minimizes errors, and improves communication between students, recruiters, and administrators. The system provides a centralized and efficient platform for managing all placement-related activities in a structured manner.

The project also highlights the importance of web technologies in solving real-world problems and improving organizational processes. In the future, the system can be enhanced by integrating advanced features such as automated notifications, resume screening, data analytics, and mobile application support. These improvements will further increase the efficiency and usability of the system, making it more robust and scalable.

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